

Angelopoulos & Bates (2022)

A Gentle Introduction to Conformal Prediction

Paper Overview

1. Why this paper exists

Modern machine-learning models return *point predictions* but tell you nothing about how much to trust each one. Bayesian posteriors, bootstrap confidence intervals, and quantile regression all give you a notion of uncertainty, but each comes with assumptions: a prior, a noise model, exchangeable resamples, or a correctly-specified quantile family. *Conformal prediction* (CP) — introduced by Vladimir Vovk and collaborators Vovk, Gammerman, and Shafer (2005) — gets you a guarantee that is *distribution-free* (works for any data distribution), *finite-sample* (works for any sample size, not just asymptotically), and *model-agnostic* (works around any predictor).

Key Insight

The promise of conformal prediction in one sentence: wrap any predictor in a procedure that turns its point output into a *set* of plausible outputs, with a hard mathematical guarantee that the true answer is inside the set at least $1 - \alpha$ of the time. You do not have to trust the model, the data distribution, or large-sample asymptotics. The guarantee only requires that your data points be *exchangeable* (the order they arrived in doesn't matter).

The Angelopoulos–Bates monograph Angelopoulos and Bates (2022)¹ is the de-facto practitioner entry point to CP. It has three layers: (i) a four-step *recipe* producing valid prediction sets around *any* model, (ii) a catalogue of nonconformity scores for classification,

¹*Foundations and Trends in Machine Learning*; arXiv:2107.07511.

regression, and full distributional output, and (iii) a survey of extensions to risk control, conditional coverage, covariate shift, and data-efficient variants. Every method ships with a Python notebook.

Plain English

Why this is the field's standard reference. Before Angelopoulos–Bates, CP literature was scattered across two communities: theoretical statisticians (Vovk–Gammerman–Shafer, Lei–Wasserman, Barber–Candès) writing in classical-statistics language, and a smaller ML-practitioner community using ad-hoc bootstrap variants. This monograph translated the theory into a recipe-card format that ML engineers could pick up in an afternoon, paired with executable notebooks. Practitioner adoption of CP roughly tripled in the two years after its release.

This overview walks through the constructions, makes the underlying intuition explicit, and points at the connections to the rest of the cp-summaries collection.

2. Background and terminology

Three terms do most of the work.

A *predictor* $\hat{f} : \mathcal{X} \rightarrow \mathcal{Y}$ is any function that has been fitted on training data — linear regression, gradient-boosted trees, a deep network. CP treats $\hat{f}$ as a *black box*: it does not inspect $\hat{f}$'s internals.

A sequence $(Z_1, \dots, Z_{n+1})$ is *exchangeable* if its joint distribution is invariant under permutation. This is *strictly weaker* than independent and identically distributed (i.i.d.): for example, draws from a Pólya urn are exchangeable but not independent. The practical implication is what matters: *any random reshuffling of an i.i.d. dataset is exchangeable*. CP's coverage guarantee needs this and nothing more.

What This Definition Means

Exchangeability in one sentence. A dataset is exchangeable if you could shuffle its rows and end up with a dataset that has the same statistical properties. The standard i.i.d. assumption is one (strong) way to get this; CP works with the weaker, more realistic version.

A *calibration set* is a held-out subset of the data, not used for training $\hat{f}$. In the split-conformal variant the calibration set produces the cutoff used to construct prediction sets. Finally, the *miscalibration level* $\alpha \in (0, 1)$ is the maximum allowed failure probability: setting $\alpha = 0.1$ asks for 90% prediction sets.

3. The four-step recipe

Given a fitted predictor $\hat{f}$, a calibration set $\{(X_i, Y_i)\}_{i=1}^n$ disjoint from $\hat{f}$'s training data, and a target miscoverage level α :

Step 1: heuristic uncertainty. Start with any $\hat{f}$. Nothing further is required of the model.

Step 2: nonconformity score. Design a function $s : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$ that quantifies “how surprising would the pair (x, y) be given $\hat{f}$?” — larger means more surprising. For regression a natural choice is the absolute residual $s(x, y) = |y - \hat{f}(x)|$; for classification, $s(x, y) = 1 - \hat{f}_y(x)$ (one minus the predicted softmax probability of the true class).

Plain English

The score is the user's design choice. The coverage guarantee holds for *any* score function. The choice you make affects the *shape* and *size* of the resulting prediction sets — for example, simple absolute residuals give constant-width intervals, while σ -scaled residuals or CQR-style scores give locally adaptive intervals. Validity is automatic; everything else is engineering judgment.

Step 3: calibration quantile. Compute the scores on the calibration data $s_i = s(X_i, Y_i)$ for $i = 1, \dots, n$, and take a specific empirical quantile:

$$\underbrace{\hat{q}}_{\substack{\text{cutoff that calibrates} \\ \text{the prediction set}}} = \text{Quantile}\left(\underbrace{\{s_i\}_{i=1}^n}_{\substack{\text{scores on} \\ \text{held-out data}}} ; \underbrace{\frac{\lceil (n+1)(1-\alpha) \rceil}{n}}_{\substack{\text{slightly above } 1-\alpha; \\ \text{the } +1 \text{ fixes a finite-}n \text{ bias}}} \right).$$

The peculiar $(n+1)$ -correction is the entire reason CP works at *finite* sample size: it accounts for the fact that the unseen test point will become the $(n+1)$ th observation.

How This Works

Why the $(n+1)$ adjustment. The conventional empirical $(1-\alpha)$ quantile of n samples slightly under-covers because it treats the test point as if it were drawn from an estimated distribution. CP instead asks: what cutoff guarantees that the future test point's score will be among the $\lceil (n+1)(1-\alpha) \rceil$ smallest scores out of all $n+1$ scores (calibration + test)? The answer is the formula above. As $n \rightarrow \infty$ the $+1$ vanishes; at small n (say $n = 100$) it's the difference between actual 89.1% coverage and the nominal 90%.

Step 4: prediction set. For a fresh input X_{n+1} , return

$$\underbrace{\mathcal{C}(X_{n+1})}_{\text{the set we hand to the user}} = \left\{ y \in \mathcal{Y} : \underbrace{s(X_{n+1}, y)}_{\substack{\text{score of candidate } y \\ \text{against } \hat{f}}} \leq \underbrace{\hat{q}}_{\text{calibrated cutoff}} \right\}.$$

The score catalogue. Step 2 is where domain knowledge enters. For classification with a deep network the choice $s(x, y) = 1 - \hat{f}_y(x)$ produces fixed-size sets; *Adaptive Prediction Sets* (APS) Sadinle, Lei, and Wasserman (2019) instead rank softmax probabilities and cumulate until they exceed $\hat{q}$, yielding sets that *grow* on hard inputs and *shrink* on easy ones. RAPS regularises APS to prevent pathological large sets on rare classes. For regression *Conformalised Quantile Regression* (CQR) Romano, Patterson, and Candès (2019) uses score

$$s(x, y) = \max\left(\underbrace{\hat{q}_{\alpha/2}(x) - y}_{\text{undershoot}}, \underbrace{y - \hat{q}_{1-\alpha/2}(x)}_{\text{overshoot}}\right),$$

yielding intervals that adapt to heteroscedasticity — their width varies locally with the conditional spread.

Key Insight

Score choice trades coverage for adaptivity. All scores produce valid $1 - \alpha$ marginal coverage. They differ in interval shape:

- Plain $|y - \hat{f}(x)|$: fixed-width intervals; under-covers in high-variance regions, over-covers in low-variance regions, but *averaged across all x* still hits $1 - \alpha$.
- σ -scaled or CQR: width varies locally; closer to *conditional* coverage when the scale model is good.
- APS / RAPS for classification: set size varies with input difficulty; easy inputs get singletons, hard inputs get larger sets.

The validity proof is identical for all four. The choice is purely about whether the user wants constant-width or adaptive intervals.

4. The coverage guarantee

THEOREM (Marginal coverage, Angelopoulos and Bates (2022)). *If $(X_i, Y_i)_{i=1}^{n+1}$ are exchangeable, then the prediction set defined above satisfies*

$$\underbrace{\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1}))}_{\substack{\text{probability the true label} \\ \text{falls inside the set}}} \in \left[\underbrace{1 - \alpha}_{\text{lower bound}}, \underbrace{1 - \alpha + \frac{1}{n+1}}_{\substack{\text{upper bound} \\ \text{(discreteness slack)}}} \right].$$

What This Result Says

The shape of the guarantee:

- It's a *lower* bound (the procedure may over-cover, never systematically under-cover).
- The upper bound $1 - \alpha + 1/(n + 1)$ comes from the fact that ranks are integers — at small n you might land a little above nominal.
- Crucially: this averages over both the test point *and* the calibration set draw. On a single fixed calibration set, the realised coverage is a random variable (see Beta corollary below).

Proof sketch. Exchangeability of $(X_i, Y_i)_{i=1}^{n+1}$ implies that the rank of the test score $s(X_{n+1}, Y_{n+1})$ among the $n + 1$ scores $\{s_1, \dots, s_n, s(X_{n+1}, Y_{n+1})\}$ is uniformly distributed on $\{1, \dots, n + 1\}$. Choosing the cutoff at the $\lceil (n + 1)(1 - \alpha) \rceil$ -th smallest calibration score exactly bounds the probability that the test rank exceeds the cutoff. The upper bound $1 - \alpha + 1/(n + 1)$ comes from the discrete nature of the rank.

Plain English

The proof in three sentences. Exchangeability means the unobserved test point is interchangeable with any of the n calibration points; therefore its score's rank among all $n + 1$ scores is uniform on $\{1, \dots, n + 1\}$. The cutoff is set so that the test rank exceeds the cutoff with probability at most α . Equivalently, the test score falls under the cutoff with probability at least $1 - \alpha$ — which is exactly the event that the test point is in the prediction set.

The coverage is *marginal*: it averages over both the test point *and* the calibration set draw. Conditional on a single calibration set, the realised coverage is itself random and follows a Beta distribution:

$$\underbrace{C}_{\substack{\text{realised coverage} \\ \text{for this cal. set}}} \sim \text{Beta}\left(\underbrace{\lceil (n + 1)(1 - \alpha) \rceil}_{\text{shape 1}}, \underbrace{n + 1 - \lceil (n + 1)(1 - \alpha) \rceil}_{\text{shape 2}}\right).$$

A rule of thumb that drops out: calibration set size $n \approx 1,000$ keeps realised coverage within $\pm 1\%$ of nominal with high probability.

Why This Matters

Why this Beta corollary matters for practice. The marginal coverage statement averages over calibration draws. In deployment you typically have *one* calibration set and live with whatever coverage that draw produced. If $n = 100$ and $\alpha = 0.1$, the

realised coverage on a single calibration set has a standard deviation of roughly 3% — so coverage can easily land at 85% or 93% instead of nominal 90%. Push n to 1,000 and the standard deviation drops to $\sim 1\%$. The $n \approx 1,000$ rule of thumb comes directly from solving for the calibration size that keeps the Beta standard deviation small enough for practical comfort.

5. Limits, extensions, and connections

The construction has three honest limitations. First, *marginal coverage* is weaker than *conditional coverage*, the property $\mathbb{P}(Y \in \mathcal{C}(X) \mid X = x) \geq 1 - \alpha$ for every x . Vovk has shown distribution-free conditional coverage is achievable only by trivial (infinite-measure) sets; CP gives you approximate substitutes (size-stratified or group-conditional coverage) but never the real thing in general. Second, the split-conformal variant wastes calibration data; the paper discusses *Full Conformal* (retraining for each candidate y , exact but expensive) and *Jackknife+* Barber et al. (2021) (leave-one-out residuals, distribution-free $1 - 2\alpha$ guarantee) as the data-efficient alternatives. Third, exchangeability breaks under distribution shift; *Weighted CP* Tibshirani et al. (2019) extends the guarantee to known covariate shift by reweighting calibration scores with likelihood ratios.

Why This Matters

The three honest limits of marginal-coverage CP.

- **Marginal $\neq$ conditional:** a 90% marginal interval may cover 99% of easy cases and 80% of hard cases. If your downstream decision depends on per-individual reliability, you need approximate-conditional variants (Mondrian, group-balanced, class-conditional).
- **Calibration data is sacrificed:** split CP leaves potential model accuracy on the table. Jackknife+ and Cross-CP recover most of it at modest extra compute.
- **Exchangeability is a strong assumption in production:** any time series, any drifting feature distribution, any selection effect can violate it. Specialised methods (ACI, EnbPI, weighted CP, NexCP) exist for the most common violation modes.

The paper also surveys *Conformal Risk Control* Angelopoulos et al. (2024) — replacing the indicator coverage loss with any bounded *monotone* loss and tuning a threshold so that the expected loss meets a budget — and *Learn-Then-Test* for non-monotone risks. These extensions are why CP keeps spreading: any time you can name a controllable risk, there is now a recipe to bound its expectation in a distribution-free way.

The other reader's guides in this collection extend the foundation in specific directions.

Dieuleveut & Zaffran (2025) give the proof architecture and coverage hierarchy in more detail. Stocker et al. (2025) organise the post-exchangeability landscape for time series into a four-family taxonomy. Xu & Xie (2023) develop EnbPI, the canonical bootstrap-ensemble method for time series. Bao et al. (2025) benchmark the regression variants head-to-head.

References

- Angelopoulos, Anastasios N. and Stephen Bates. 2022. “A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification.” *Foundations and Trends in Machine Learning*. ArXiv:2107.07511.
- Angelopoulos, Anastasios N. Stephen Bates, Adam Fisch, Lihua Lei, and Tal Schuster. 2024. “Conformal Risk Control.” *International Conference on Learning Representations*.
- Barber, Rina Foygel, Emmanuel J. Candès, Aaditya Ramdas, and Ryan J. Tibshirani. 2021. “Predictive Inference with the Jackknife+.” *Annals of Statistics*.
- Romano, Yaniv, Evan Patterson, and Emmanuel Candès. 2019. “Conformalized Quantile Regression.” In *Advances in Neural Information Processing Systems*.
- Sadinle, Mauricio, Jing Lei, and Larry Wasserman. 2019. “Least Ambiguous Set-Valued Classifiers with Bounded Error Levels.” *Journal of the American Statistical Association*.
- Tibshirani, Ryan J. Rina Foygel Barber, Emmanuel J. Candès, and Aaditya Ramdas. 2019. “Conformal Prediction Under Covariate Shift.” In *Advances in Neural Information Processing Systems*.
- Vovk, Vladimir, Alexander Gammerman, and Glenn Shafer. 2005. *Algorithmic Learning in a Random World*. Springer.